bowl’s center, it would finally rendezvous with its dinner, as in Figure 9.1b.

But if the bowl were shaped differently, as in Figure 9.1c, things would turn out differently. Imagine again that the bowl starts out very hot and that the worm pile still lies at the bowl’s center, now high up on the central bump. Were you to drop the frog in, it would again wildly jump this way and that, remaining oblivious to the prize perched on the central plateau. Then, as the bowl cooled, the frog would again settle itself, reduce its jumping, and slide down the bowl’s smooth sides. But because of the new shape, the frog would never make it to the bowl’s center. Instead, it would slide down into the bowl’s valley and remain at a distance from the worm pile, as in Figure 9.1d.



Figure 9.1 **(**a**)** A frog dropped into a hot metal bowl incessantly jumps around. **(**b**)** When the bowl cools, the frog calms down, jumps much less, and slides down to the bowl’s middle.

If we imagine that the distance between the frog and the worm pile represents the value of a field—the farther the frog is from the worms, the larger the value of the field—and the height of the frog represents the energy contained in that field value—the higher up on the bowl the frog happens to be, the more energy the field contains—then these examples convey well the behavior of fields as the universe cools. When the universe is hot, fields jump wildly from value to value, much as the frog jumps from place to place in the bowl. As the universe cools, fields “calm down,” they jump less often and less frantically, and their values slide downward to lower energy.